

PENGYU WANG

Updated: August 28, 2026

4801 24th Ave NE, Seattle, WA 98105
(+1) 608-886-8792 | pwang425@uw.edu

EDUCATION

University of Washington, Michael G. Foster School of Business <i>Ph.D. in Operations Management</i> Advisors: Prof. Shi Chen, Prof. Yong-Pin Zhou, Prof. Masha Shunko	Seattle, WA, USA Jan. 2023 – expected 2027
University of Washington, Department of Economics <i>Ph.D. in Economics</i> (program; transitioned to OM)	Seattle, WA, USA Sept. 2020 – Dec. 2022
University of Wisconsin–Madison, Department of Economics <i>Master of Science in Economics</i>	Madison, WI, USA Aug. 2018 – June 2020
Western University <i>Bachelor of Arts, Honors Specialization in Economics</i>	London, Ontario, Canada Sept. 2014 – June 2018

RESEARCH INTERESTS

Topics: Sustainable operations, Platform operations, AI infrastructure operations, Global supply chain challenges, Semi-conductor industry and supply chain.

Methods: Stochastic optimization, Game theory, Data analysis & Causal inferences, Mean-field games and control, Reinforcement learning.

RESEARCH

(†: Co-first authors).

Working Papers

1. **Wang, P.**, Chen, S., Hou, X. (2026). *Collective ESG Investment: Mitigation, Greenwashing, and Dynamic Strategic Interactions*. Reject & Resubmit at *Manufacturing & Service Operations Management* (Apr, 2026), Resubmitted (Aug, 2026).
 - Presented at INFORMS 2025.
 - *Scheduled presentation at INFORMS 2026.
 - We build a dynamic stochastic framework to study how heterogeneous firms make collective ESG investment decisions, including greenwashing and substantive mitigation, when their actions are strategically interdependent and interact with an ESG-aware financial market.
2. Li, Y.†, **Wang, P.**†, Zhou, Y., Shunko, M. *Can Less Be More? A Hybrid Solution to Print-on-Demand Platforms*. Reject & Resubmit at *Manufacturing & Service Operations Management* (Nov, 2025)

- Accepted by and presented at MSOM conference 2024.
 - Studies hybrid make-to-stock / make-to-order (MTS/MTO) production and assortment regulation on print-on-demand platforms. Develops analytical models and contracts to align incentives between platforms and content creators.
3. **Wang, P.[†]**, Zhang, J.[†], Chen, S., Yang, M. (2026). *Platform-Empowered Digital Finance and Traditional Investing: Spillover Effects through Financial Literacy and Risk Tolerance*. Under Review at *Manufacturing & Service Operations Management* (Aug, 2026).
 - *Accepted by and scheduled talk at CIST conference 2026, and DSI conference 2026.
 - This paper studies whether digital finance substitutes for traditional channels or broadens market participation. Using individual-level investment data, we conceptualize digital finance as a gateway that reduces participation frictions, facilitates experiential learning, and promotes migration into adjacent financial markets.
 4. **Wang, P.**, Zhou, Y., Shunko, M. *Marketplace Governance and Production Strategy: Decentralized Assortment and Hybrid Fulfillment in Creator Economy*. In preparation for submission to *Management Science*.
 - * Scheduled talk at DSI 2026 conferece.
 - We study how creator portfolios are formed in digital marketplaces and the resulting production challenges for platforms. The separation between assortment creation and production decisions creates a need for marketplace governance, which platforms can address through market instruments that reshape creator portfolios and improve performance.

Work in Progress

5. **Wang, P.**, Chen, S. (2026) *A Two-Layer Hybrid Graph Mean-Field Game for Distributed LLM Inference Demand*.
6. **Wang, P.**, Chen, S. (2026) *Global Supply Chain Networks, Dynamics, and Disruption*.
7. **Wang, P.**, Zhang, J., Chen, S. (2026). *Who Are the Winners? Business Opportunity Management in Business Relationships*.

Proceedings

8. **Wang, P.** (2017) *The Causal Relationship between Socioeconomic Status and Smoking Behavior: Evidence from Canada*. Proceedings in Advances in Economics, Business & Management.
-

CONFERENCES & INVITED TALKS

Decision Science Institute 2026, San Francisco, CA (scheduled)
 INFORMS Annual Meeting 2026, San Francisco, CA (scheduled)
 MOSM 2026 Annual Meeting, Boston, MA
 POMS Annual Meeting 2026, Reno, NV

POMS-HK 2026, Shenzhen, China
INFORMS Annual Meeting 2025, Atlanta, GA
SICS Annual Meeting 2025, Berkeley, CA
INFORMS Annual Meeting 2024, Seattle, WA
INFORMS MOSM 2024 Annual Meeting 2024, Twin-City, MN
INFORMS Annual Meeting 2023, Pheonix, AZ

Society Services

Session chairs: INFORMS 2024, MSOM 2024, DSI 2026, INFORMS 2026

TEACHING EXPERIENCE

Invited Instructor: (Evaluations out of 5.0).

Winter 2026, IGB 250: 4.5

Summer 2026, IGB 250: 4.6

Teaching Assistance:

Autumn 2023, SCM 530A: 4.6; SCM 513: 4.5.

Winter 2024, BUS AN 516A: 4.2; BUS AN 516B: 4.1.

Spring 2024, OPMGT 520A: 4.3

Winter 2025, EMBA 533A: 4.5

Spring 2025, QMETH 505A: 4.1; SCM 520A: 4.2

REFERENCES

Prof. Shi Chen

Professor of Operations Management, David and Dana Lewis Endowed Professor

Foster School of Business, University of Washington

shichen@uw.edu

PhD Advisor

Prof. Yong-Pin Zhou

Professor of Operations Management, Longbrake Endowed Professor of Operations Management

Foster School of Business, University of Washington

yongpin@uw.edu

PhD Advisor

Prof. Masha Shunko

Associate Professor of Operations Management

Foster School of Business, University of Washington

mshunko@uw.edu

PhD Advisor